INCH-POUND

MIL-M-14/1
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, PHENOLIC, THERMOSETTING, CONTAINING CELLULOSE FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Phenolic resin molding compounds, cellulose filled, shall be of the following types, as specified:

Type CFG	–	Cellulose filler, general-purpose
Type CFI-5	–	Cellulose filler, impact-resistant; nominal impact strength, 0.5 foot-pound per inch notch
Type CFI-10	–	Cellulose filler, impact-resistant; nominal impact strength, 1.0 foot-pound per inch notch
Type CFI-20	–	Cellulose filler, impact-resistant; nominal impact strength, 2.0 foot-pounds per inch notch
Type CFI-30	–	Cellulose filler, impact-resistant; nominal impact strength, 3.0 foot-pounds per inch notch
Type CFI-40	–	Cellulose filler, impact resistant; nominal impact strength, 4.0 foot-pounds per inch notch.

REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

AMSC N/A FSC 9330

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Type CGF. This type is a general-purpose, wood-flour-filled phenolic compound.

Type CFI-5. This type is a moderate-impact, cotton- or paper-filled phenolic compound.

Type CFI-10. This type is a medium-impact, cotton rag-filled phenolic compound.

Type CFI-20. This type is a high-impact, rag- or cotton-filled phenolic compound.

Type CFI-30. This type is a high-impact, cotton-filled phenolic compound.

Type CFI-40. This type is the highest-impact grade of cotton-filled phenolic compound.

TABLE I. *Qualification test requirements for phenolic resin molding compounds, cellulose filled.*

Requirement	Type CFG	Type CFI-5	Type CFI-10	Type CFI-20	Type CFI-30	Type CFI-40
Mechanical/physical						
Compressive strength, endwise	25,000	23,000	20,000	20,000	19,000	18,000
Flexural strength	9,000	8,000	8,000	8,000	8,000	8,000
Heat deflection temperature	115	115	115	115	115	115
Heat resistance	115	115	115	115	115	115
Impact strength, side[1]	0.2	0.5	1.0	2.0	3.0	4.0
Tensile strength	6,000	5,700	5,700	5,700	5,700	6,000
Water absorption	3.0	4.0	4.0	4.0	4.0	4.0
Electrical						
Dielectric breakdown:						
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	30	18	18	18	18	18
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	2.5	2.5	2.5	2.5	2.5	2.5
Dielectric strength:						
Short-time test	300	250	240	210	250	175
Step-by-step test	200	150	180	150	150	–
Short-time test	75	50	40	45	20	25
Step-by-step test	45	27.5	27.5	25	10	15
Combustion						
Flame resistance:						
Ignition time	60	60	60	60	60	60
Burning time	270	330	330	330	330	330

[1]The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. *Batch acceptance test requirements for phenolic resin molding compounds, cellulose filled.*

Property to be tested	Type CFG	Type CFI-5	Type CFI-10	Type CFI-20	Type CFI-30	Type CFI-40
Arc resistance	–	–	–	–	–	–
Dielectric strength, step-by-step	45	27.5	27.5	25	10	15
Flexural strength	9,000	8,000	8,000	8,000	8,000	8,000
Impact strength, side[1]	0.2	0.5	1.0	2.0	3.0	4.0
Water absorption	3.0	4.0	4.0	4.0	4.0	4.0

[1]The side of the test specimen is that area formed by the chase of the mold.

Custodians:
Army – ME
Navy – SH
Air Force – 20

Review activities:
Army – ER, MI, MR, AR, EA
Navy – AS, OS
DLA – GS

User activities:
Navy – YD, EC

Preparing activity:
Navy - SH
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